MCPA Cheat Sheet

Model Context Protocol Associate exam prep — MCP 2026-07-28 spec

LF certification discount codes
promo.kubernetes lab.dev

Revision tags (2026-07-28): New added · Changed behavior changed · Removed dropped · Dep. Deprecated (discouraged, still works)

Decision rules

- Tools: the LLM decides. Resources: the host decides. Prompts: the user decides
- A request gets a response with the same `id` (unless cancelled); a notification gets none
- Malformed request: JSON-RPC error. Tool execution error or bad tool arguments: `isError` result
- Cancel on stdio with `notifications/cancelled`; on Streamable HTTP, close the response stream
- From 2026-07-28: no handshake or session; every request carries version and capabilities in `_meta`
- Unless the server is trusted, distrust its tool descriptions and annotations; validate all tool results before the LLM sees them
- Servers check token audience and never forward the token upstream (no passthrough)
- Secrets such as passwords or API keys go through URL mode, never form mode

Tool call flow

1. `tools/list` returns tool definitions
2. Host sends the user's question and tool definitions to the LLM
3. LLM picks a tool and arguments
4. Host asks the user to confirm, if needed
5. Client sends `tools/call`; server runs the tool
6. Result (`content`, `isError`) goes back to the LLM, which writes the answer

01 MCP basics

M×N integration problem	One-off app-to-tool integrations (M×N) become M+N with a shared protocol
host	AI app; connects the LLM and clients, handles consent and policy
client	Lives in the host; 1:1 connection to one server
server	Provides tools, resources and prompts; runs tools
LLM	Picks the tool and arguments; does not run it
data layer	Inner layer: JSON-RPC messages, primitives, notifications
transport layer	Outer layer: connections, delivery, authorization
primitive	Unit of functionality (tools, resources, prompts, ...)
capability	Declared feature support; undeclared means unusable
`initialize` Removed	Old version and capability handshake
session Removed	Per-connection state (`Mcp-Session-Id`)
stateless New	Each request carries everything it needs
`_meta` Changed	Carries version and capabilities on every request
handle New	Server-issued state ID passed as a tool argument (non-normative guidance)

02 JSON-RPC (JSON Remote Procedure Call)

request	Has `id` and `method`; answered with the same `id`
response	Same `id`; holds `result` or `error`, never both
notification	No `id`; no reply
id	String or integer, never null
`resultType` New	`complete` or `input_required`
Parse error, Invalid Request, Internal error	Bad JSON, malformed request, server failure
Method not found	Unknown method
Invalid params Changed	Bad params, unknown tool, missing resource (not bad tool arguments)
MCP-defined errors New	`UnsupportedProtocolVersion`, `MissingRequiredClientCapability`, `HeaderMismatch`

03 Server primitives and methods

tools	Functions the LLM decides to call
resources	Data (URIs) the host decides to use
prompts	Templates the user picks
`server/discover` New	Supported versions and capabilities (servers must implement)
`tools/list`	Lists tools; resources and prompts also use `*/list`
`tools/call`	Runs a tool by name with arguments
`inputSchema`	JSON Schema for arguments
`outputSchema`	Schema for `structuredContent`
`isError`	Tells the LLM the tool execution failed
annotation	`readOnlyHint`, `destructiveHint`, `idempotentHint`, `openWorldHint`; all hints, not guarantees

03 Server primitives and methods (cont.)

content block	text, image, audio, resource_link, resource
`resources/read` Changed	Reads text or base64 blob by URI; missing resource returns Invalid params
`resources/templates/list`	Lists URI templates such as `file:///{path}`
`prompts/get`	Returns prompt messages for the given arguments
`completion/complete`	Autocomplete for prompt and resource template arguments

04 Client features

elicitation	Server asks the user for more input
form mode	Flat fields from `requestedSchema`; no secrets
URL mode	Collects secrets, third-party OAuth or payment on an external page (not for MCP authorization)
`action`	`accept`, `decline` or `cancel`
`notifications/elicitation/complete` Removed	URL mode completion notification
sampling Dep.	Server asks the client-side LLM to generate a response
`includeContext` Dep.	Omit or `none`, not `thisServer` / `allServers`
roots Dep.	Directory hints, not access control
`notifications/roots/list_changed` Removed	Roots changed notification

05 Utilities and flow control

MRTR New	*Multi Round-Trip Requests.* Server asks via `inputRequests`; client resends the original request with `inputResponses` and a new `id`
`input_required` New	Result asking for more input
`requestState` New	Echoed back as is; the server validates it
`progressToken`	Put in `_meta` to get progress notifications
`notifications/progress`	`progress` only goes up
`notifications/cancelled` Changed	Cancel on stdio; servers send it only to end a `subscriptions/listen` stream
pagination	`cursor` and `nextCursor`; server sets page size
`subscriptions/listen` New	Long-lived stream for list changes and resource updates
`list_changed` Changed	List changed notification, only if requested via `subscriptions/listen`
`resources/subscribe` Removed	Per-URI subscription
`ttlMs`, `cacheScope` New	Cache lifetime and sharing scope
logging Dep.	Server sends logs at the `logLevel` set in request `_meta`
`ping` Removed	Liveness check

06 Transports and headers

stdio	Subprocess; stdout for messages, stderr for logs
Streamable HTTP	One endpoint, one POST per message (notification: 202)
HTTP+SSE Dep.	Old HTTP transport (2024-11-05)
`MCP-Protocol-Version` Changed	Must match the version in `_meta`
`Mcp-Method` New	Method name, for routing
`Mcp-Name` New	Tool or prompt name, or resource URI
`x-mcp-header` New	Schema property that mirrors a tool argument into `Mcp-Param-{name}`
`Mcp-Session-Id` Removed	Session ID
`Last-Event-ID` Removed	Used to resume a broken SSE stream
GET stream Removed	Old endpoint for server-initiated messages (405 recommended)
`Origin` **check**	Blocks DNS rebinding; bad Origin gets 403

07 Authorization (OAuth)

OAuth 2.1	Optional; HTTP should use it, stdio should use environment credentials
resource server	The MCP server's OAuth role
authorization server	Issues tokens
Authorization Server Metadata	Lists the authorization server endpoints
PKCE	*Proof Key for Code Exchange.* Stops authorization code theft; clients must use it, with `S256` whenever capable

07 Authorization (OAuth) (cont.)

bearer token	In the `Authorization` header on every request, never in the query string
audience	Server checks the token was issued for it (required)
`resource` **parameter**	Names the target server URI (required for clients)
Protected Resource Metadata	Points to the authorization server
scope step-up	On 403 `insufficient_scope`, reauthorize with the required scopes added to the old ones
`iss` **check** New	Defends against mix-up attacks
CIMD	*Client ID Metadata Documents.* HTTPS URL as `client_id` (recommended)
DCR Dep.	*Dynamic Client Registration.* On-the-fly client registration

08 Threats and controls

token passthrough	Forwarding a received token upstream (forbidden)
prompt injection	Instructions in external input that change behavior
tool poisoning	Instructions hidden in tool names or descriptions
tool shadowing	A malicious description manipulates how another server's tool is used
rug pull	Tool definition changes after approval
confused deputy	Proxy forwards to a third-party authorization server without per-client consent
SSRF	*Server-Side Request Forgery.* Malicious server lists internal URLs in OAuth metadata for the client to fetch
state handle hijacking New	Stolen or guessed handle reaches another user's state; bind handles to users
local server compromise	Malicious startup command in client config; show the full command and get consent
authorization URL validation	Block `javascript:` and similar authorization URLs; allow only http(s), never open via a shell
least privilege	Minimal permissions and scopes; step up only when needed
human-in-the-loop	A person can review and deny tool calls
consent	Explicit permission for data access and tool calls
sandboxing	Isolation that limits damage
input validation	Check arguments before the tool runs (required for servers)
rate limiting	Cap on calls (required for servers)
audit log	Who did what; no credentials in logs

09 Ecosystem

MCP Inspector	Official debugger that calls a server directly
MCP Registry	Official list of public server metadata
MCPB	*MCP Bundle.* Local server packaged as one .mcpb file
gateway	Routing, auth and policy in front of many servers
extension	Optional feature, declared in the `extensions` field of capabilities
Tasks Changed	Extension for polling long-running work
MCP Apps	Extension serving UI through `ui://` resources

10 Foundation and related protocols

AAIF	*Agentic AI Foundation.* Linux Foundation home of MCP, goose and AGENTS.md
SEP	*Specification Enhancement Proposal.* Spec change proposal
Core Maintainers	Approve SEPs, decide removals
feature lifecycle New	Active, Deprecated (min. 12 months, or min. 90 days for security risks) and Removed
A2A	*Agent2Agent.* Protocol for agents to hand off work
OpenTelemetry New	Tracing via `traceparent` in `_meta`

11 General agent terms

agent	LLM program that pursues a goal with tools
ReAct	*Reasoning and Acting.* Alternates reasoning and actions
chain-of-thought	Step-by-step intermediate reasoning
orchestration	Coordinates LLMs, tools and agents
function calling	LLM outputs a function name and arguments as structured data
RAG	*Retrieval-Augmented Generation.* Retrieves documents into context before answering

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